K-Means Clustering in Image Compression

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As modern media increases in bitrate and resolution, more burden is placed on server storage and network infrastructure to carry large files to end users. Raw images often contain millions of pixels, and the high preservation of detail in this data leads to large file sizes, which are inefficient for storage and transmission. If we compressed these large images by "reducing the number of unique colors in an image" while preserving visual fidelity, end users could still receive comparable images while placing far less strain on the internet, and decreasing the required rate of server storage expansion (Alzaber et all, 2018).

Traditional compression techniques like JPEG rely on frequency-domain transformations. K-means clustering is a robust unsupervised learning algorithm that can reduce image complexity by grouping similar pixel values. This approach involves "clustering the pixels in an image into a smaller number of groups and then representing each group by its mean color" (GeeksforGeeks, 2025). An unsupervised algorithm is advantageous in this scenario because no data prep is required, a necessity in today's exponentially growing media landscape.

K-means clustering reduces color space complexity. Instead of storing every unique pixel color, K-means identifies k representative colors (centroids) and assigns each pixel to the nearest one (Wan, 2019). By grouping similar colors, the algorithm maintains the image's overall appearance while significantly reducing data. K-means is computationally efficient and well-suited for large datasets like high-resolution images, and it provides simple compression level control by adjusting k.

References

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